

Quicksort

Full Tree Trace

Note this is not how the algorithm calls recursively. It is for intuition on how it works.

For this see the quicksort recursive call trace

[4,3,2,7,9,8,6,1,5]

[4,3,2,7,9,8,6,1,5]



[3,2,1] + [4] + [7,9,8,6,5]

Divide

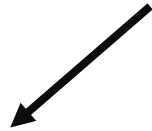
1. Pick 4 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]



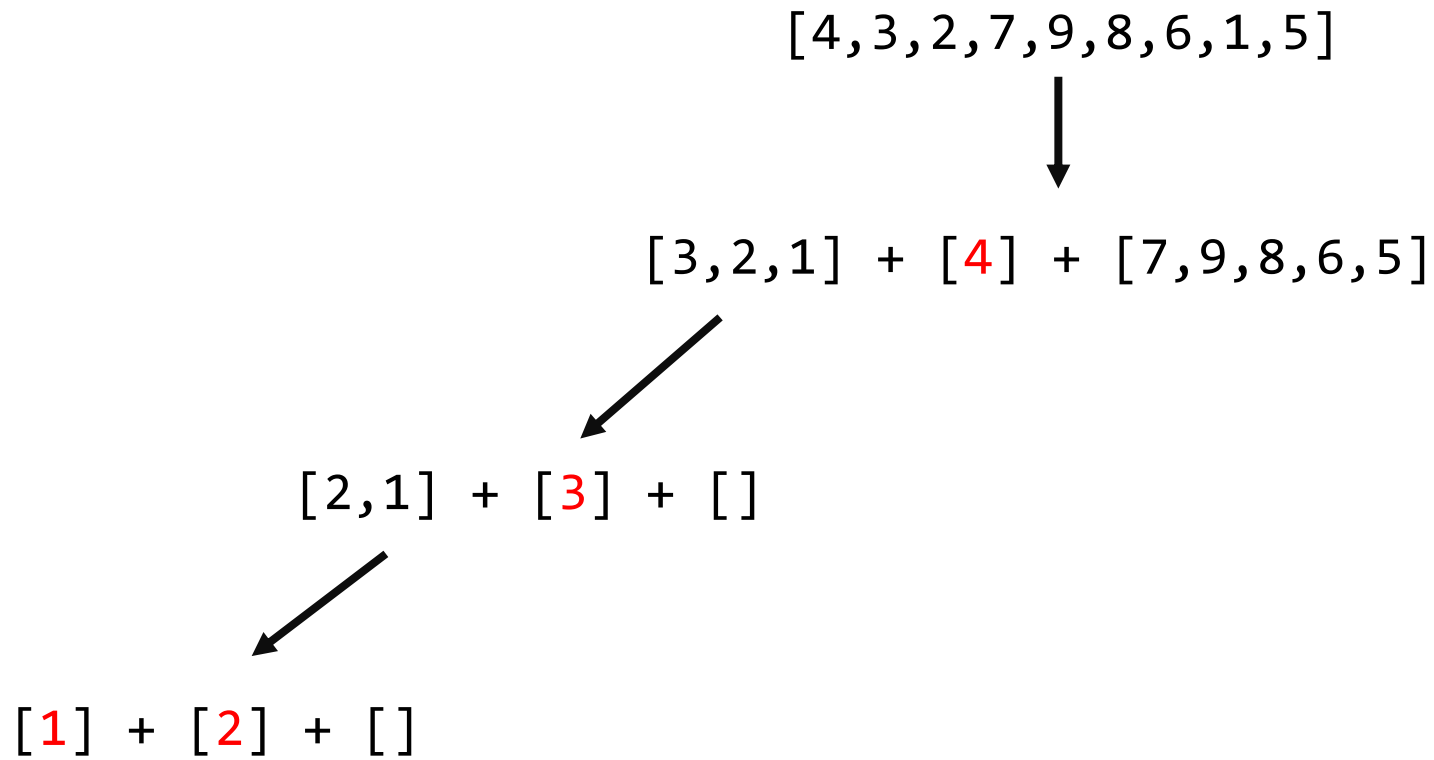
[3,2,1] + [4] + [7,9,8,6,5]



[2,1] + [3] + []

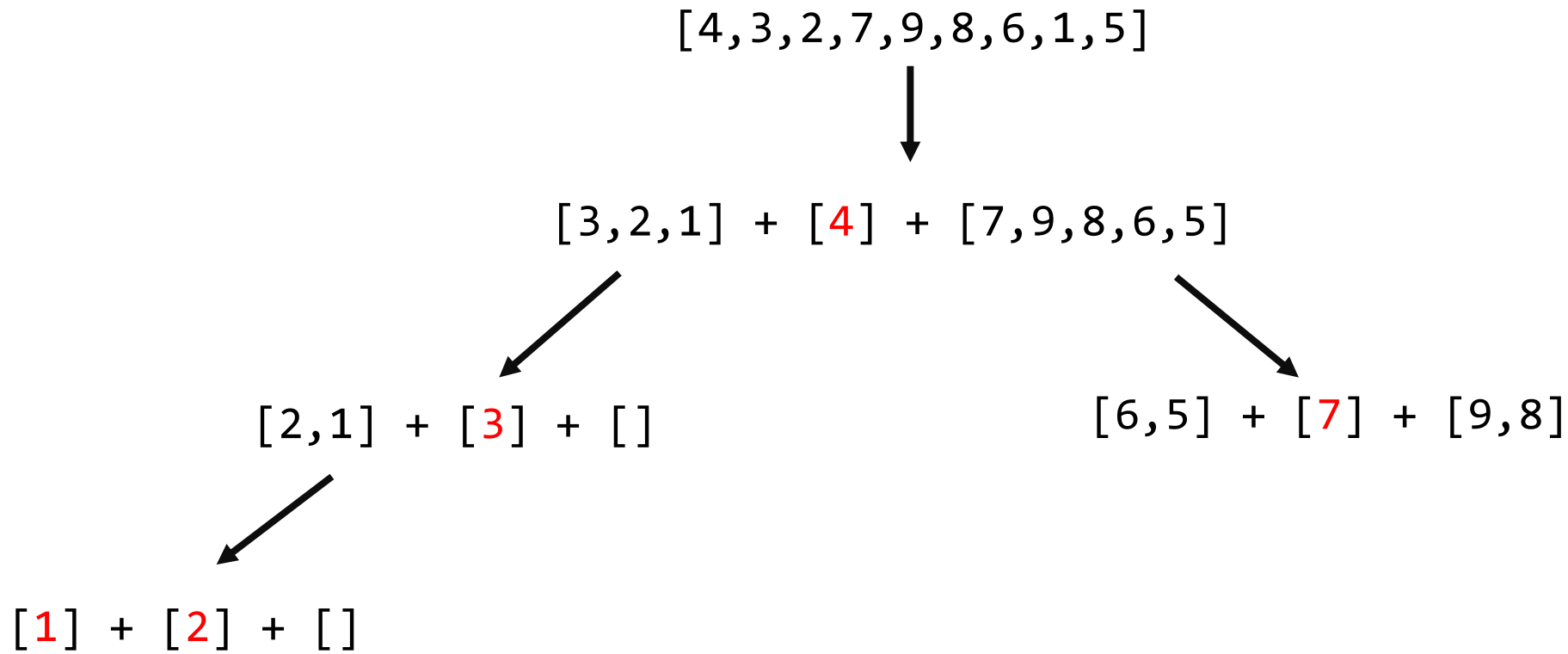
1. Pick 3 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide



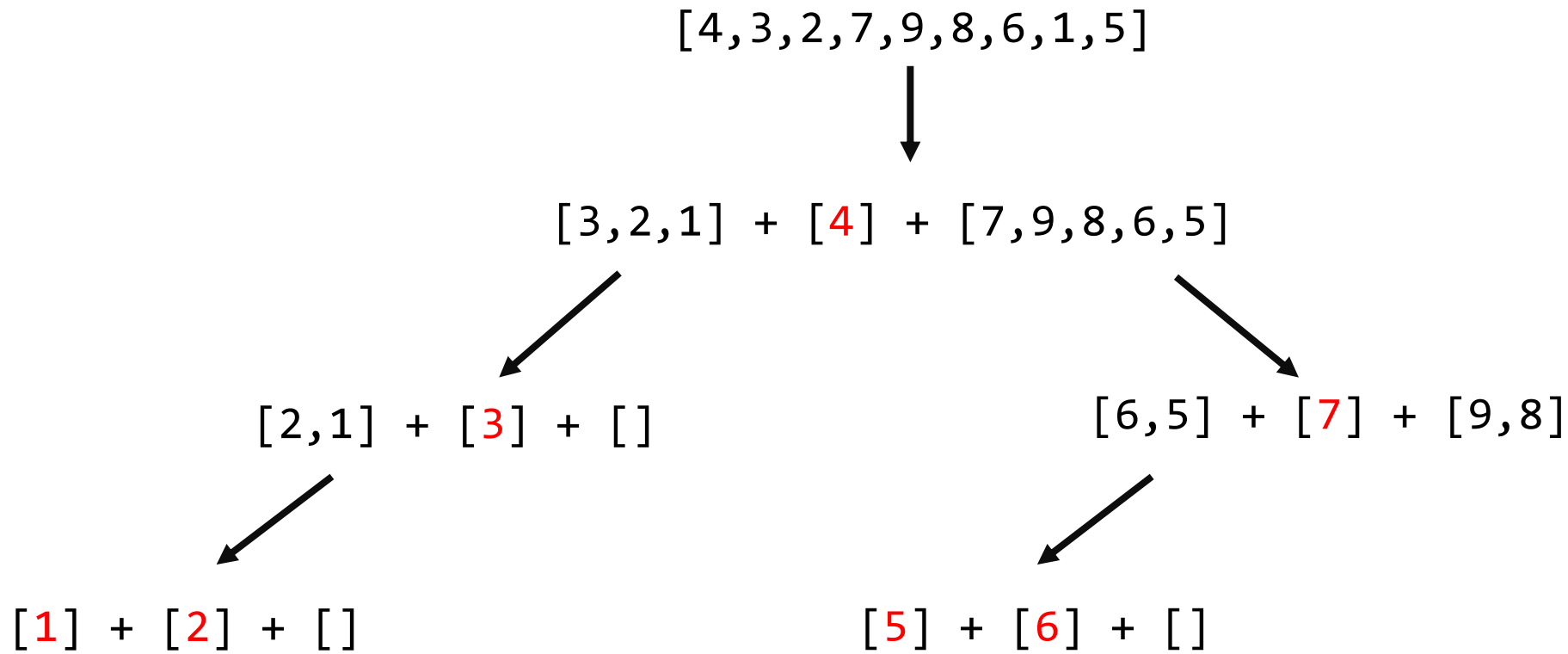
1. Pick 2 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide



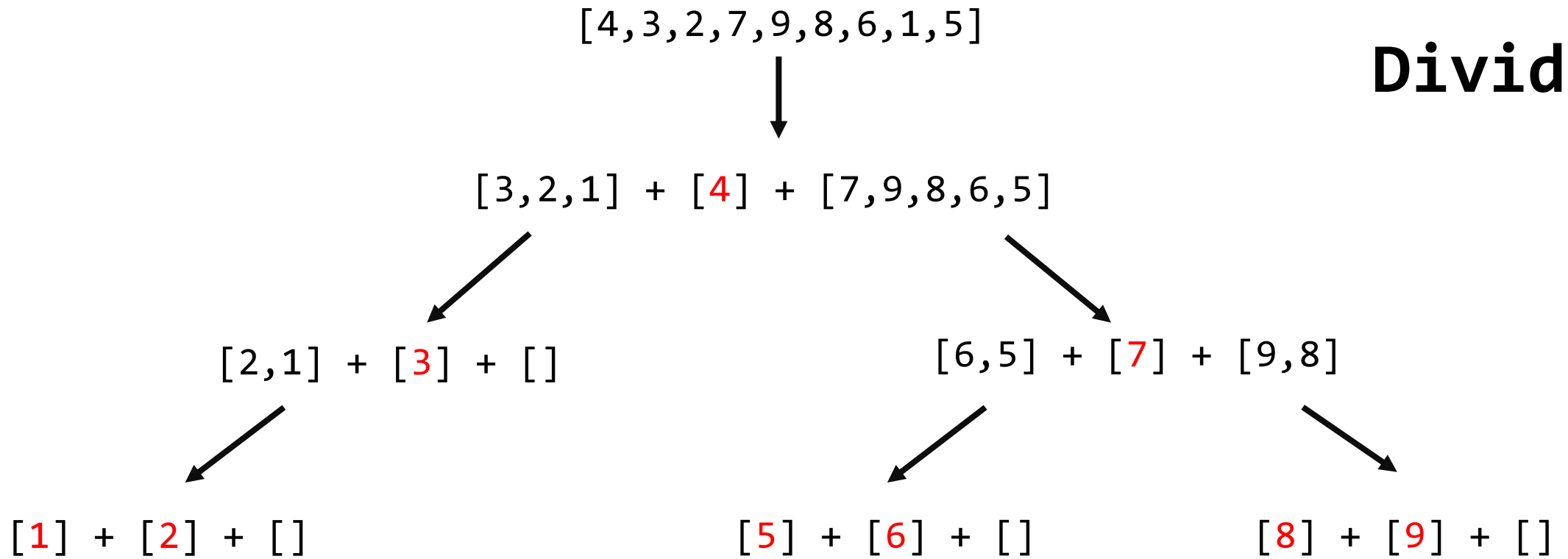
1. Pick 7 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

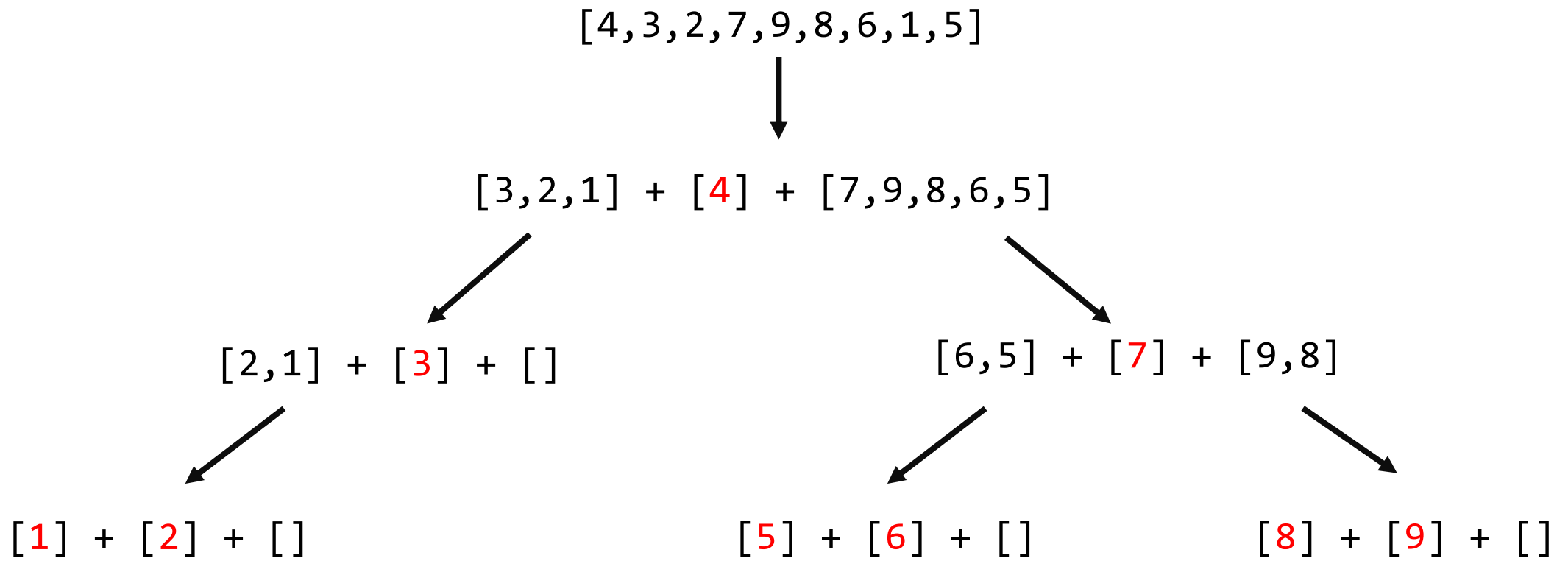


1. Pick 6 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide



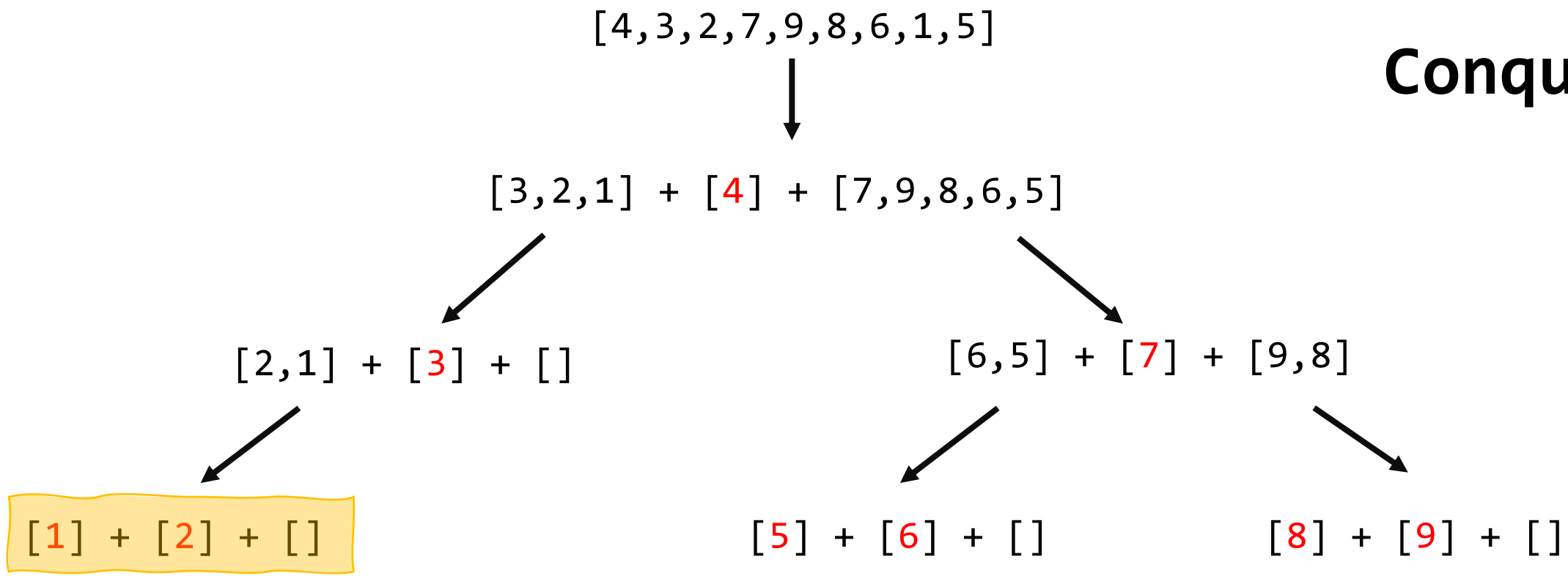
1. Pick 9 as the pivot
2. Place items lower to the left
3. Place items higher to the right



Notice that all the pivots (in red) are
in sorted order from left to right!

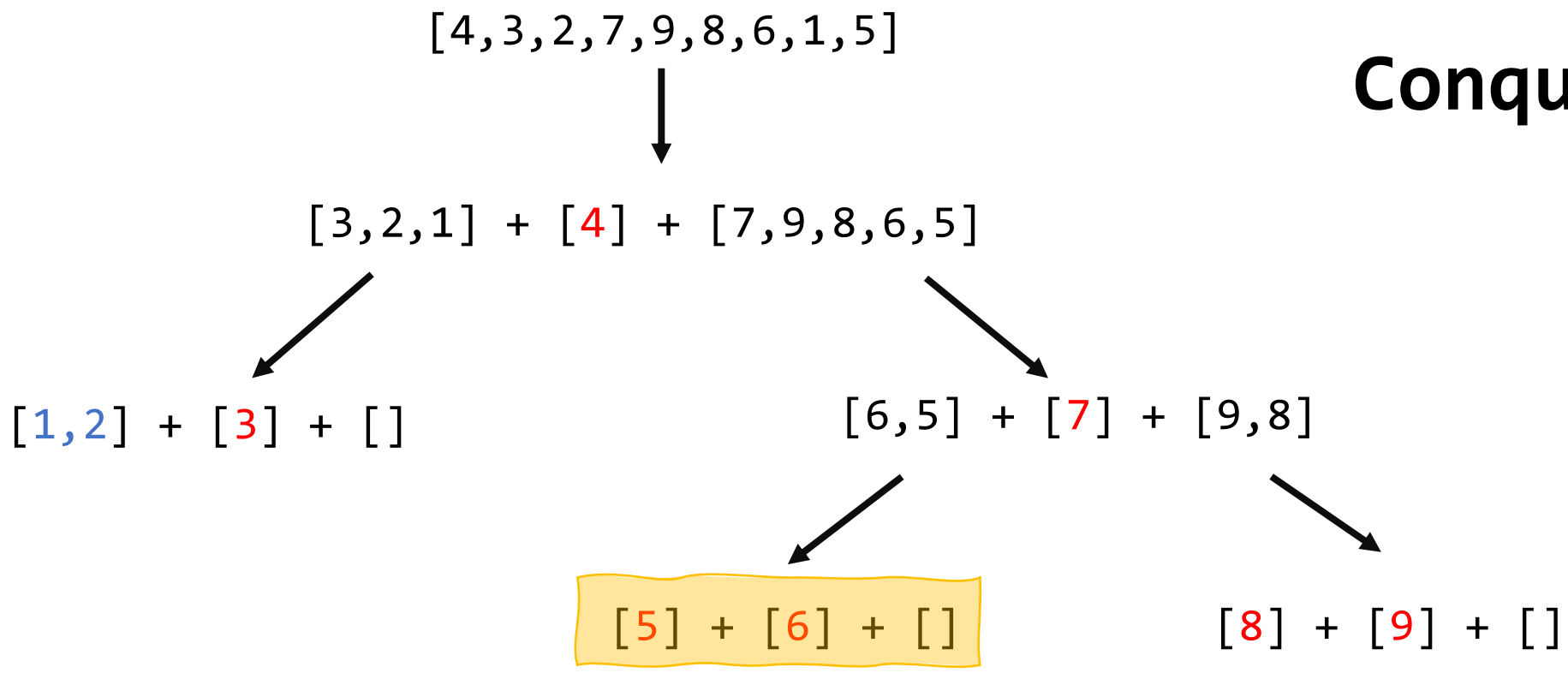
These are then combined back together

Conquer



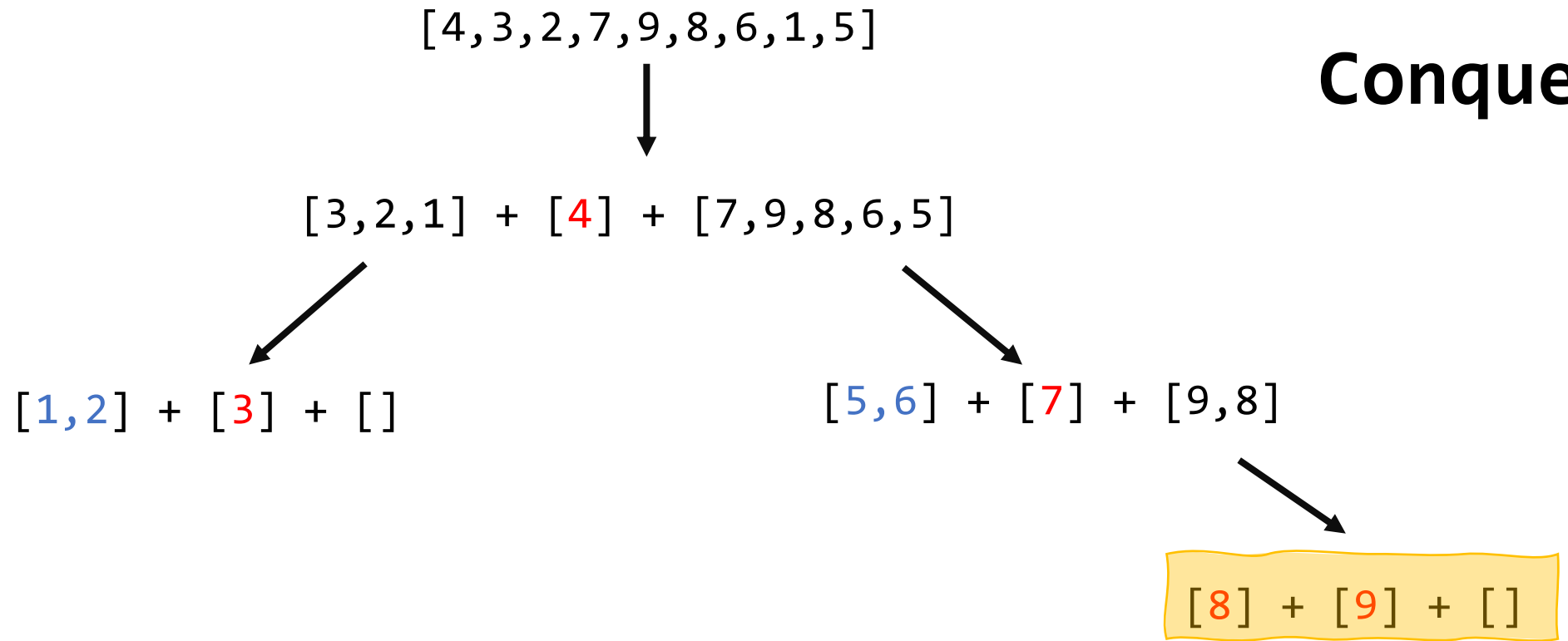
Combine the lists

Conquer



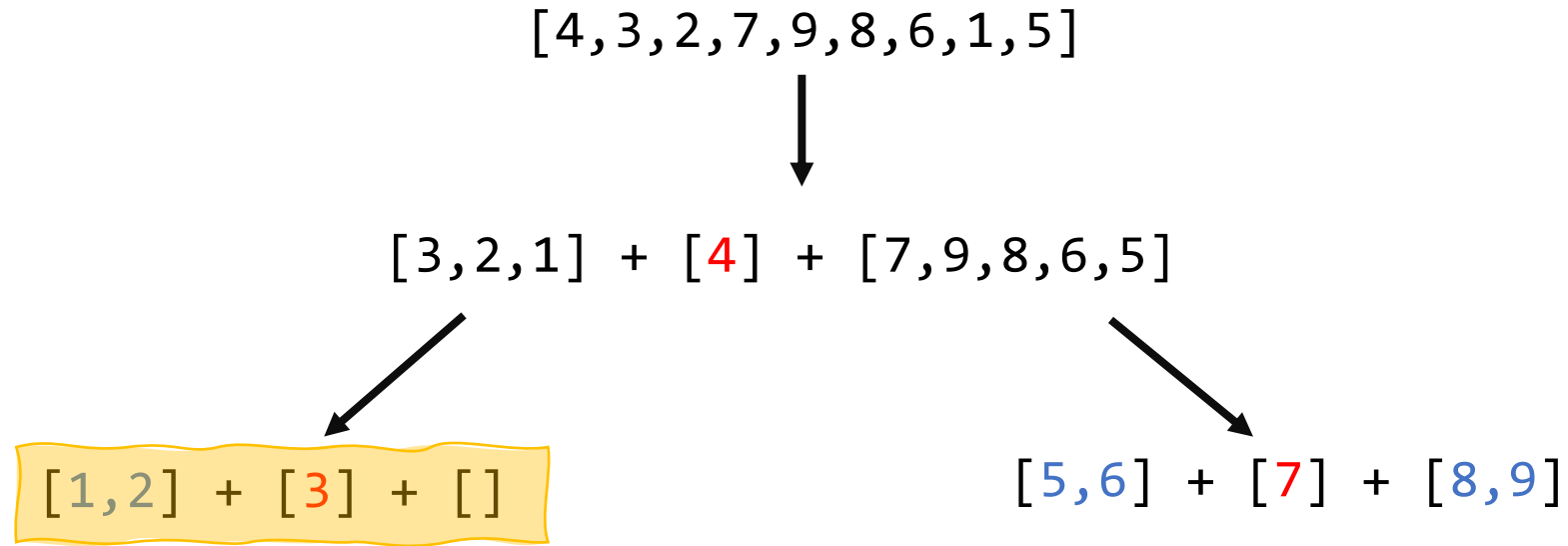
Combine the lists

Conquer



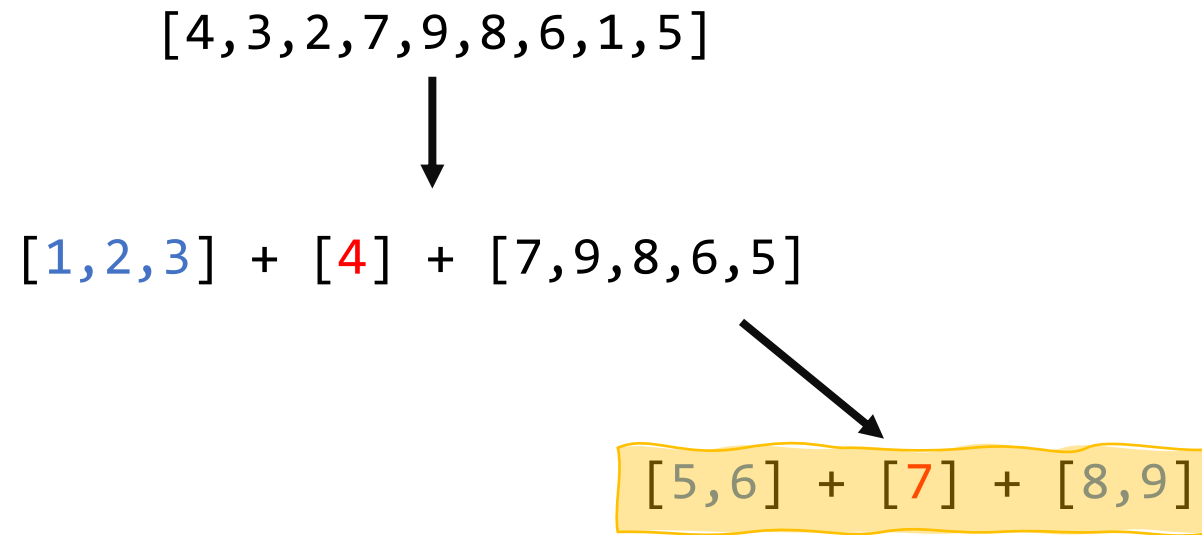
Combine the lists

Conquer



Combine the lists

Conquer



Combine the lists

[4,3,2,7,9,8,6,1,5]



[1,2,3] + [4] + [5,6,7,8,9]

Conquer

Combine the lists

[1,2,3,4,5,6,7,8,9]

Original list is now sorted